Actuation-Aware Sensor Validation for Closed-Loop Hydroponic Systems: An Auditable Supervisory Interface

Abstract

This paper proposes AASVR-R, a streaming measurement-to-actuation reliability layer for closed-loop hydroponic systems. Unlike a conventional detector, AASVR-R can record a suspect measurement while withholding its authority to drive an actuator. The novelty is an auditable supervisor interface: validation, trusted-state update, rectification, authorization, and response verification are separate decisions, while robust gates and Page/CUSUM residual tests provide signal-domain evidence. Held-out replay on a real NFT hydroponic deployment shows detection parity with the strongest tuned baselines and a lower replay false-authorized-actuation rate, but this reduction is coupled to a higher missed-authorization rate and should not be read as measured closed-loop control improvement. Plausible stuck-at detection remains at or below chance without actuator-state logs or independent reference sensing, identifying the main requirement for future deployment evidence.

Keywords: sensor validation, data rectification, fault detection, process control, hydroponics, actuation

1. Introduction

Hydroponic controlled-environment systems depend on continuous measurement and actuation. In a recirculating Nutrient Film Technique (NFT) installation, the streams that describe pH, electrical conductivity (EC), air temperature, humidity, water temperature, water level, and carbon dioxide (CO2) can also trigger dosing pumps, chillers, valves, cooling devices, and circulation equipment. A faulty measurement is therefore not only a data-quality issue; it can become a physical command that perturbs the nutrient solution or growth environment.

Standard filtering and anomaly-detection methods usually score the signal. Closed-loop operation also requires a second question: whether the current value is trustworthy enough to update the controller state or authorize an actuator. A single high-pH spike should not authorize acid dosing, and a temporary EC excursion should not be treated the same as a persistent concentration shift. This motivates actuation-aware sensor validation, where logging, state update, rectification, and actuation authorization are separate decisions.

We present a closed-loop NFT hydroponic platform and propose AASVR-R, an actuation- and response-aware sensor-validation layer, as its fault-resilience mechanism. The contribution is structural: AASVR-R formalizes a supervisory measurement-to-actuation interface in which robust gates decide whether a sample is plausible, a state machine decides whether it can update the trusted state, a rectifier decides what value is exposed to control, an authorization rule decides whether actuation is allowed, and a response residual checks whether the measured process reacts as expected after an authorized command. The contributions are:

1. an integrated hydroponic deployment dataset and sensor-to-actuator formulation for NFT control;
2. AASVR-R, a streaming supervisory interface that couples standard validation evidence, bounded rectification, risk-aware authorization, and response-reliability memory;
3. a held-out synthetic-fault protocol with validation-only tuning, tuned baselines, ablations, component analysis, and paired statistical tests;
4. a reproducible Docker and LaTeX workflow that regenerates the tables, figures, and diagnostic outputs from the data and code available in the project repository.

2. Related Work

2.1. Sensor faults in closed-loop systems

Sensor data-quality reviews distinguish missingness, outliers, drift, stuck values, saturation, and inconsistent rates as recurring faults in deployed systems [1]. Agricultural IoT systems face the same problem when sensor failure or miscalibration propagates into monitoring or control decisions [2, 3]. In closed-loop hydroponics, these faults have downstream consequences because an erroneous measurement can change the actuator command. This motivates treating validation, rectification, and actuation as separate decisions rather than as one smoothing step.

2.2. Filtering and monitoring baselines

The baseline family used in this paper covers common online filters, process-monitoring methods, model-based statistical tests, and lightweight one-class anomaly detectors. Hampel-style robust outlier detection uses robust scale estimates to suppress isolated deviations [4]. Kalman filtering gives a probabilistic recursive estimator for linear dynamical systems [5]. EWMA and CUSUM-style monitoring are standard approaches for persistent shifts [6, 7]. Generalized likelihood ratio (GLR) tests provide a classical statistical framework for abrupt change detection [8]. Analytical redundancy, parity/observer residuals, and reconfiguration logic are central themes in fault-detection and isolation (FDI) for control systems [9, 10, 11]; AASVR-R does not replace these model-based methods, but supplies the controller-facing supervisor that decides whether a measurement or residual may update trusted state, authorize actuation, or only enter the audit log. Its distinction from classical reconfiguration is practical rather than algorithmic: it exposes the accept/hold/authorize/verify decisions that a small hydroponic controller can execute under limited instrumentation. PCA and recursive PCA-style residual monitoring use residual statistics to capture coordinated changes or adaptive operating regimes [12, 13, 14]. Isolation Forest, One-Class SVM, and local outlier factor add common one-class machine-learning comparisons for anomaly detection [15, 16, 17]. Existing hydroponic automation studies have demonstrated reliable pH/EC regulation and IoT monitoring, but they generally treat validation as part of measurement filtering or rule execution rather than as a separate authorization decision [18, 19]. AASVR-R is complementary to these methods because it adds a control-facing authorization interface.

2.3. External process datasets

Public process datasets are useful stress tests for measurement validation, but their labels do not necessarily represent hydroponic sensor faults. HAI 21.03 contains industrial-control attack data from a hardware-in-the-loop testbed [20], and SKAB contains multivariate process-loop experiments with anomaly and changepoint labels [21]. These datasets are retained in the reproducibility artifacts as supplementary transfer checks, not as headline evidence. SWaT, WaDi, and Tennessee Eastman remain relevant larger benchmarks for future validation under their respective access and redistribution constraints [22, 23, 24].

3. Hydroponic System Design and Deployment

The target platform is a closed NFT hydroponic system for lettuce growth. NFT systems recirculate a shallow nutrient stream through sloped growth channels, allowing water and nutrient reuse when solution management is reliable [25, 26]. The deployed system is organized into two physical modules. The Growth Module houses the plants, lighting, air-temperature and humidity sensing, CO2 sensing, visual monitoring, and environmental actuation. The Nutrient Solution Container Module stores and recirculates the nutrient solution and contains the water-quality sensors and nutrient-solution actuators.

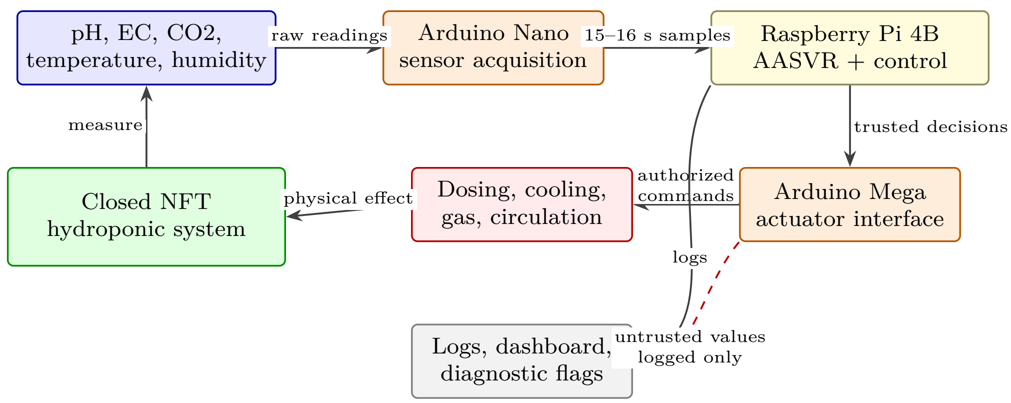

Figure 1: Hydroponic system architecture and AASVR-R location in the sensing-to-actuation path.

The Growth Module uses a DHT22 sensor for air temperature and humidity and an MH-Z19C nondispersive infrared CO2 sensor for gas concentration. CO2 regulation is implemented through a normally closed pneumatic solenoid valve connected to a CO2 cylinder. Air-temperature regulation is supported by a thermoelectric cooler. Two 40 W LED grow-light arrays provide a controlled light cycle, and a camera provides visual monitoring of the crop during the growth cycle.

The Nutrient Solution Container Module uses an Atlas Scientific analog pH kit, an Atlas Scientific Mini Conductivity K 1.0 kit for EC, a water-temperature sensor, and a JSN-SR04T waterproof ultrasonic distance sensor for water level. pH correction is implemented through acid/base dosing, and EC correction through concentrated nutrient solution or distilled-water addition. After a pH or EC dosing action, the dosing subsystem is disabled for a 10-minute mixing interval before further dosing is allowed. A refrigerative water chiller regulates nutrient-solution temperature, an aerator runs throughout the growth cycle to support dissolved oxygen, and a water pump recirculates nutrient solution through the NFT channels.

The control unit connects sensors to an Arduino Nano and actuators to an Arduino Mega. Both microcontrollers communicate with a Raspberry Pi 4 Model B, which serves as the central control computer. This sensor-microcontroller-edge-computer structure is consistent with prior hydroponic automation systems, but the present paper focuses on the measurement-to-actuation reliability layer rather than on dashboarding as a contribution [18, 19]. The Raspberry Pi receives median-filtered sensor readings from the Arduino Nano, logs the readings, updates the ThingSpeak-backed dashboard, applies the validation/control logic, and sends authorized actuation commands to the Arduino Mega. The dashboard and camera feed are retained as deployment infrastructure; they are not claimed as the research contribution of this paper.

Table 1: Hydroponic platform components relevant to measurement validation and actuation. Water level is listed because it is part of the deployed system, but it is excluded from the primary quantitative analysis because the available raw values are not defensibly calibrated.

Subsystem	Measurements	Actuation or use
Growth module	Air temperature, humidity	Thermoelectric cooling
Growth module	CO2 concentration	CO2 solenoid valve
Growth module	Camera feed	Visual crop monitoring
Growth module	Light schedule	LED grow lights
Solution module	pH	Acid/base dosing, 10 min lockout
Solution module	EC	Nutrient/water dosing, 10 min lockout
Solution module	Water temperature	Refrigerative chiller
Solution module	Water level	Warning only in this analysis
Solution module	Recirculation state	Pump and aerator operation

The evaluation below treats this system as a deployed measurement-to-actuation stream: raw measurements are replayed through validation methods, transformed into trusted states, and then passed through an authorization rule before any actuator command is counted.

4. Problem Formulation

For hydroponic sensor s at time t, let $y_s(t)$ be the raw measurement and $\hat{x}_s(t)$ be the trusted value exposed to control. The method observes recent history $Y_s(t-w:t)$, broad physical limits $[P_{s,\min}, P_{s,\max}]$, a control band $[L_s^a, U_s^a]$, a rate limit r_s, and any available actuator state $a(t)$. In this system, s may correspond to pH, EC, humidity, air temperature, water temperature, or CO2, while $a(t)$ may represent dosing, cooling, circulation, or gas-release activity.

We define a measurement as *untrusted* when it is missing, physically implausible, rate-inconsistent, inconsistent with recent trusted state, or inconsistent with expected actuator effects. In replay, we define a *false-authorized actuation* as an authorization caused by untrusted evidence. A *missed authorization* is a failure to authorize action when the trusted state remains outside the control band.

The validation decision is separated from the control decision. Let $g_s(t) \in \{0,1\}$ denote whether a raw sample passes all measurement gates, and let $z_s(t)$ denote one of five states: normal, suspect transient, persistent deviation, accepted regime shift, and fault alert. The controller receives $\hat{x}_s(t)$ and $A_s(t)$, not the raw value alone. This separation is necessary because a value can be physically possible but still too uncertain to authorize actuation.

The conceptual evaluation objective is

$$\begin{aligned} J(\theta) =& w_1 E_{\text{pass}} + w_2 E_{\text{reject}} + w_3 A_{\text{false}} + w_4 A_{\text{missed}} \\ &+ w_5 D_{\text{delay}} + w_6 T_{\text{unsafe}} + w_7 C_{\text{compute}}, \end{aligned} \tag{1}$$

where the terms respectively measure anomalous pass-through, false rejection, false-authorized actuation, missed authorization, detection delay, residence outside safe bands, and computational cost. The validation sweep uses the operative scalar

$$\begin{aligned} J_{\text{val}}(\theta) =& \text{BA}(\theta) - 0.01 A_{\text{false}}(\theta) - 0.05 R_{\text{missed}}(\theta) \\ &- 0.02 R_{\text{unsafe}}(\theta) - 0.0005 F_{\text{alarm}}(\theta) - 0.001 D_{\text{delay}}(\theta), \end{aligned} \tag{2}$$

where BA is balanced accuracy, A_{false} is the mean false-authorized actuation count, R_{missed} is missed authorization divided by unsafe-band opportunities, R_{unsafe} is the unsafe-band sample fraction, F_{alarm} is the mean false-alarm event count, and D_{delay} is mean detection delay in samples. Runtime remains a logged diagnostic rather than a validation-selection term.

5. Proposed Method: AASVR-R

5.1. Overview

Figure 2 summarizes the AASVR-R pipeline. The method separates five decisions that are often conflated: measurement validation, trusted-state update, rectification, actuation authorization, and response verification. This separation is the core design choice. In the hydroponic system, this means that a suspicious pH value can be stored with a diagnostic flag while acid/base dosing remains suppressed until the value is persistent and trusted, and repeated dosing authority can be removed if the pH stream does not respond after a command.

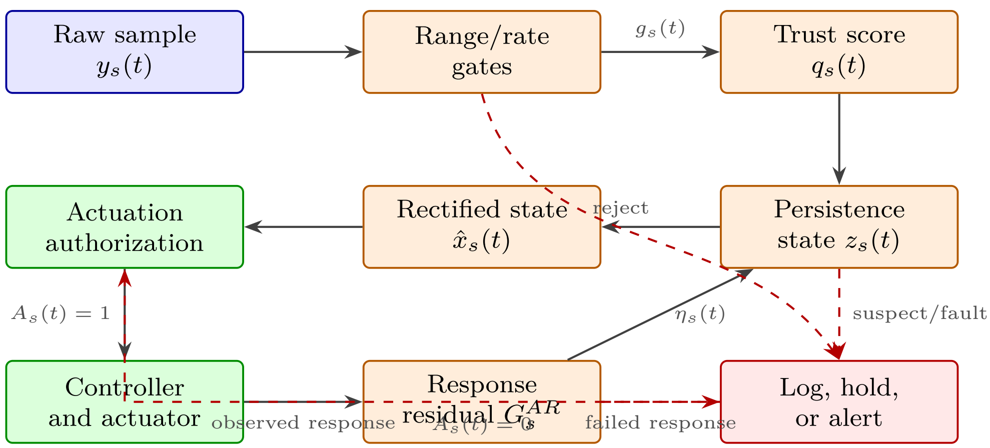

Figure 2: AASVR-R architecture. Measurement validation, trusted-state update, rectification, actuation authorization, and response verification are separate decisions.

5.2. Robust plausibility gate

For each sensor, AASVR-R estimates a robust scale over adjacent changes,

$$\sigma_{\Delta,s}(t) = 1.4826\,\mathrm{MAD}\left(\Delta y_s(t-w:t)\right). \tag{3}$$

The accepted deviation tolerance is

$$\xi_s(t) = \max(\xi_{s,\min}, c_s\sigma_{\Delta,s}(t), r_s\Delta t, u_s). \tag{4}$$

A reading must satisfy broad physical range, maximum rate, and trusted-state deviation gates. The three elementary gates are

$$\begin{aligned} G_s^P(t) &= \mathbf{1}\{P_{s,\min} \le y_s(t) \le P_{s,\max}\}, \\ G_s^R(t) &= \mathbf{1}\{|y_s(t) - y_s(t-1)|/\Delta t \le r_s\}, \\ G_s^D(t) &= \mathbf{1}\{|y_s(t) - \hat{x}_s(t-1)| \le \xi_s(t)\}. \end{aligned} \tag{5}$$

A separate trend guard $G_s^T(t)$ rejects slow monotonic movement when no relevant actuator activity explains the direction of change. A stuck-value guard $G_s^S(t)$ rejects sustained exact repetition after a preceding change, which targets frozen plausible readings without treating ordinary low-variance noise as a fault. To make the drift component less dependent on a single-window monotonicity rule, AASVR-R also uses a two-sided CUSUM residual test. For residual $e_s(t) = y_s(t) - \hat{x}_s(t-1)$,

$$\begin{aligned} C_s^+(t) &= \max\{0, C_s^+(t-1) + e_s(t) - \kappa_s(t)\}, \\ C_s^-(t) &= \max\{0, C_s^-(t-1) - e_s(t) - \kappa_s(t)\}, \end{aligned} \tag{6}$$

where $\kappa_s(t) = \alpha_c \xi_s(t)$ is the allowed drift allowance. The sequential residual gate is

$$G_s^C(t) = \mathbf{1}\{\max(C_s^+(t), C_s^-(t)) \le h_c \xi_s(t)\}. \tag{7}$$

This is the Page/CUSUM change-detection principle applied to the trusted-state residual: isolated residuals decay, while persistent one-sided departures accumulate until they are no longer trusted for state update. The validation-selected hydroponic configuration uses $\alpha_c = 0.10$ and $h_c = 0.60$ for the primary sensors. The overall gate is therefore

$$g_s(t) = G_s^P(t) G_s^R(t) G_s^D(t) G_s^T(t) G_s^S(t) G_s^C(t) G_s^M(t), \tag{8}$$

where $G_s^M(t)$ is one only when the value is present and parseable. The broad physical range is deliberately different from the crop-control band: for example, a pH value can be undesirable for lettuce while still being physically possible and therefore not automatically a sensor fault.

5.3. Trust score

AASVR-R also records a continuous trust score in $[0, 1]$ from physical-range, rate, persistence, actuator-consistency, and missingness components. Trusted-deviation, uncommanded-trend, stuck-value, CUSUM-residual, and actuator-response-residual failures cap the score through the gate outcome rather than adding another free weight. The selected replay configuration uses the score primarily as an audit variable: component analysis shows that robust gates plus trust produce the same held-out decisions as robust gates plus hold until coupled with the full persistence and authorization logic. Numerical settings are separated from the method definition and selected through the validation protocol in Section 6.4.

Table 2: Manuscript AASVR-R configuration and exhaustive sensitivity ranges. The exhaustive profile is implemented as a shardable reproducibility protocol; the manuscript tables use the held-out benchmark unless explicitly marked as exhaustive sensitivity.

Parameter family	Manuscript setting	Exhaustive sensitivity range
Robust scale c_s	6.0	2, 3, 4, 5, 6, 8, 10, 12
Trust threshold $q_{\min}$	0.70	0.50, 0.60, 0.70, 0.80, 0.90
Transient / persistent limits	2 / 5 samples	transient 1, 2, 3, 5; persistent 3, 5, 8, 12
Authorization persistence k_s	3 samples	1, 2, 3, 5, 8
Replay cooldown	4 samples	0, 1, 2, 4, 8, 16, 40
Rectification branch	hold	hold, bounded prediction
CUSUM (α_c, h_c)	(0.10, 0.60)	$\alpha_c \in [0, 0.80]$, $h_c \in [0.20, 5.00]$
Response memory λ	0.50	0.50, 0.70, 0.80, 0.90, 0.95
Risk thresholds	0.75 / 0.65 / 0.50	high 0.60–0.90, medium 0.50–0.80, low 0.30–0.60
Response window	sensor-specific 6–8	4, 6, 8, 12, 20 samples

5.4. State-machine persistence logic

AASVR-R uses five persistence states: normal, suspect transient, persistent deviation, accepted regime shift, and fault alert. A single failed gate moves a sensor into a suspect transient state. Repeated failures create a persistent-deviation state. Consistent evidence can be accepted as a new process regime, while missing, stuck, saturated, or contradictory readings escalate to fault alert. This matters for hydroponics because real mixing and thermal responses can take multiple samples after dosing or cooling, whereas isolated spikes should not immediately change the trusted state.

The state update can be written as

$$z_s(t) = \mathcal{T}\left(z_s(t-1), g_s(t), q_s(t), h_s(t), a(t)\right), \tag{9}$$

where $h_s(t)$ summarizes persistence-window evidence. This representation makes the state machine explicit without treating every transition as a separate hand-tuned rule in the manuscript. The transition function $\mathcal{T}$ is implemented by Algorithm 1.

5.5. Rectification and actuation authorization

When a reading is accepted, AASVR-R sets $\hat{x}_s(t) = y_s(t)$. When it is suspect, the method either holds $\hat{x}_s(t-1)$ or applies a bounded update limited by $r_s\Delta t$:

$$\hat{x}_s(t) = \begin{cases} y_s(t), & z_s(t) \in \{\text{normal}, \text{accepted}\}, \\ \hat{x}_s(t-1), & z_s(t) = \text{suspect}, \\ \hat{x}_s(t-1) + \text{clip}(\Delta y_s, -r_s\Delta t, r_s\Delta t), & \text{bounded}, \\ \text{alert}, & z_s(t) = \text{fault}. \end{cases} \tag{10}$$

Actuation is authorized only if the trusted value is outside the control band, the violation persists for the configured number of samples, the sensor state is normal or accepted-regime-shift, the trust score is sufficient, the actuator is not in cooldown, and the recent response reliability $\eta_s(t)$ is above the risk-specific threshold:

$$\begin{aligned} A_s(t) = &\mathbf{1}\{\hat{x}_s(t) \notin [L_s^a, U_s^a]\}\mathbf{1}\{p_s(t) \geq k_s\}\mathbf{1}\{q_s(t) \geq q_{\min}\} \\ &\times \mathbf{1}\{\eta_s(t) \geq \eta_{\min,\rho(s)}\}\mathbf{1}\{z_s(t) \in \mathcal{Z}_{\text{safe}}\}\mathbf{1}\{\ell_s(t) = 0\}. \end{aligned} \tag{11}$$

Here $p_s(t)$ is the number of consecutive trusted control-band violations, k_s is the persistence requirement, $\rho(s)$ is the action-risk class, $\mathcal{Z}_{\text{safe}} = \{\text{normal}, \text{accepted}\}$, and $\ell_s(t)$ is the actuator lockout indicator. For pH and EC, this rule complements the 10-minute dosing lockout already present in the deployed system; for CO2 and temperature, it prevents transient readings from becoming immediate gas-release or cooling commands.

5.6. Actuator-response residual

After an authorization at time t_0, AASVR-R checks whether the trusted measurement moves in the expected direction within a response window. Let $d_s(t_0) \in \{-1, 0, +1\}$ denote the expected direction: high pH or EC implies a decreasing response after acid or dilution, low pH or EC implies an increasing response after base or nutrient dosing, and cooling implies decreasing temperature. If $d_s(t_0) = 0$, the response gate is not evaluated and no response-reliability update is made for that episode. Otherwise, the response evidence is

$$M_s(t_0) = \max_{\tau_{\min} \leq \tau \leq \tau_{\max}} d_s(t_0)\left[\hat{x}_s(t_0 + \tau) - \hat{x}_s(t_0)\right]. \tag{12}$$

The response gate is

$$G_s^{AR}(t_0) = \mathbf{1}\{M_s(t_0) \geq \rho_s\}. \tag{13}$$

If this gate fails, the method logs an actuator-response residual, withholds further authority from the sensor according to the reliability memory, and may escalate to fault alert. The reliability memory is

$$\eta_s(t+1) = \lambda\eta_s(t) + (1-\lambda)c_s(t), \tag{14}$$

where $c_s(t) = 1$ after a confirmed response and $c_s(t) = 0$ after a failed response. This term is intentionally different from $q_s(t)$: $q_s(t)$ scores the current sample, while $\eta_s(t)$ scores whether recent actuation episodes produced credible sensor response.

Bounded-update observation. Assume the true process state satisfies $|x_s(t) - x_s(t-1)| \leq r_s\Delta t$ and the previous trusted estimate has error $|\hat{x}_s(t-1) - x_s(t-1)| \leq e_{t-1}$. Under the bounded rectification branch in Eq. (10), $|\hat{x}_s(t) - x_s(t)| \leq e_{t-1} + 2r_s\Delta t$ for any raw value $y_s(t)$, including arbitrarily large excursions. The result follows from applying a clipped update of magnitude at most $r_s\Delta t$ while the true state also moves by at most $r_s\Delta t$. Under a sustained suspect state, this per-sample bound can accumulate linearly, which is the formal counterpart of the empirical drift weakness. This is therefore a bounded-input statement for the controller-facing estimate, not a full closed-loop stability proof.

Algorithm 1 Streaming AASVR-R update for one sensor

Require: Raw value $y_s(t)$, previous trusted state $\hat{x}_s(t-1)$, recent history, limits, actuator state.

1: Estimate robust local scale $\sigma_{\Delta,s}(t)$ and tolerance $\xi_s(t)$.
2: Apply physical-range, rate, trusted-deviation, missingness, trend, and stuck-value gates.
3: If actuator response is pending, check the response-window residual and update $\eta_s(t)$.
4: Update the persistence state from the gate outcome.
5: **if** the reading is accepted **then**
6: Set $\hat{x}_s(t) \leftarrow y_s(t)$.
7: **else if** the reading is transient suspect **then**
8: Hold or bounded-predict $\hat{x}_s(t)$.
9: **else**
10: Keep the last safe estimate and log a fault-alert decision.
11: **end if**
12: Authorize actuation only if trusted state, persistence, score, reliability, and cooldown rules are satisfied.
13: **return** Trusted value, sensor state, trust score, actuation flag, and log record.

6. Evaluation Protocol

6.1. Hydroponic deployment data

The hydroponic data contain 121244 raw rows collected from 2024-02-27 to 2024-03-28 at a median cadence of approximately 16 seconds. The primary analysis uses EC, pH, humidity, air temperature, water temperature, and CO2. Water level is part of the deployed system but is excluded from the primary analysis because the available raw values are not defensibly calibrated.

6.2. Agronomic context

Agronomic measurements are retained only as deployment context. The per-plant harvest file contains 63 records across two experiments and three treatments, but only P1 produced the sensor-to-actuator decision stream used by AASVR-R. The analysis therefore does not claim a causal crop-yield effect; it uses the crop cycle to show that the measurement-to-actuation pipeline operated during a real deployment.

6.3. Supplementary external stress tests

HAI 21.03 and SKAB are retained in the reproducibility outputs as supplementary stress tests rather than as main evidence. Their labels describe cyber-physical attacks and process-loop anomalies, not hydroponic sensor faults. The direct-transfer results are therefore reported only as a limitation check.

6.4. Fault injection protocol

Synthetic faults are injected into windows sampled from the real hydroponic background. For each sensor, fault type, magnitude, and duration, three deterministic replicates are generated and assigned to tune, validation, and test splits. The reported hydroponic benchmark uses only the held-out test split; AASVR-R and baseline hyperparameters are selected on validation trials. Fault windows include 90 pre-fault samples and 120 post-fault samples so that detection delay and false alarms are evaluated around the injected event rather than in isolation.

To address sensitivity to fault-design choices, the repository also defines a shardable exhaustive profile that extends the compact protocol to additional stuck, saturation, nonlinear drift, gain-drift, and intermittent-burst cases. The present manuscript reports the compact held-out benchmark because it is fully executed, validation-separated, and aligned with the available hydroponic deployment evidence; the larger profile is retained as reproducibility infrastructure rather than a completed result.

Table 3: Compact synthetic fault-injection protocol for the current hydroponic replay benchmark. The repository also defines a larger shardable stress-test profile, but those full-factorial results are not used as manuscript evidence.

Fault type	Injection rule over fault window $\mathcal{I}$	Purpose
Spike	$y'_s(t) = y_s(t) + m$	Single-window offset
Multi-spike	$y'_s(t) = y_s(t) + b_t m,\ b_t \in \{-1, 1\}$	Alternating impulse faults
Dropout	$y'_s(t) = \mathrm{NaN}$	Missing samples
Drift	$y'_s(t) = y_s(t) + m(t - t_0)/\lvert\mathcal{I}\rvert$	Slow monotonic deviation
Bias	$y'_s(t) = y_s(t) + m$	Persistent offset
Noise burst	$y'_s(t) = y_s(t) + \epsilon_t,\ \epsilon_t \sim \mathcal{N}(0, m^2)$	Increased variance
Step	$y'_s(t) = y_s(t) + m$	Abrupt sustained change
Stuck-at	$y'_s(t) = y_s(t_0)$	Constant repeated value
Saturation	$y'_s(t) = m$	Sensor rail or implausible fixed value

6.5. Counterfactual response replay

A second replay evaluates the actuator-response residual. Replay-derived controller authorizations are treated as pseudo-actuator commands because the available feed does not contain real actuator-state columns. Healthy and response-fault trials are implementation checks for the response mechanism, not measured physical closed-loop disturbances. We do not infer actuator commands from the same sensor thresholds used for validation, because doing so would make the actuator evidence circular.

6.6. Prospective limitation-closing evidence

The reproducibility package now treats three missing evidence sources as explicit optional schemas rather than informal future work. First, a real or reconstructed-from-controller actuator-state log should record timestamped actuator identity, commanded and measured relay state, command source, target sensor, expected response direction, dose or runtime, lockout status, and manual override status. The reconstruction utility accepts independent controller, relay, dashboard, or ThingSpeak command exports, but it does not infer commands from sensor threshold crossings. When such records are supplied, the pipeline reports measured command-state agreement and directional post-command response, converting the response residual from counterfactual replay into deployment evidence for those episodes. Second, an independent reference-measurement file should pair raw sensor values with a calibrated handheld or benchtop reference instrument, including device identity and operator notes. This supports direct scoring of plausible bias, drift, and stuck-at readings. Third, water-level reintegration requires paired raw ultrasonic readings and physical reference levels in centimeters. The current reproducibility utility fits a linear calibration, $\ell_{\mathrm{cm}} = ay_{\mathrm{raw}} + b$, and reports slope, intercept, RMSE, and R^2; water level should remain excluded until those diagnostics are acceptable for the physical tank geometry. Missing optional files are allowed in the current pipeline and produce absence-status tables; they do not change the held-out hydroponic benchmark.

6.7. Baselines and metrics

The baselines are raw thresholding, moving average, moving median, Hampel, Kalman, EWMA, CUSUM, PCA-style monitoring, GLR, recursive PCA, Isolation Forest, One-Class SVM, and local outlier factor. A duplicated physical-threshold rule remains in the reproducibility outputs but is not interpreted as a separate scientific baseline. Every displayed

Table 4: Evidence status for the current revision. The table separates available deployment evidence from prospective evidence needed to convert replay checks into measured closed-loop actuator-response claims.

Evidence source	Current status	Current use	Claim boundary
Hydroponic sensor feed	Available	Replay and synthetic-fault background	No independent metrology truth
Synthetic fault labels	Available	Controlled held-out scoring	Injected faults are not all hardware faults
Actuator command/state log	Not retained	Not used as measured evidence	No physical actuator-response claim
Independent reference measurements	Not continuous	Not used for full scoring	Plausible bias/drift truth unavailable
Water-level calibration	Not available	Water level excluded	No water-level fault claim

baseline is tuned on the validation split using Eq. (2) and then wrapped in the same persistence/cooldown authorization supervisor for replay false-authorized-actuation scoring. The hydroponic synthetic-fault benchmark reports precision, recall, specificity, balanced accuracy, detection delay, false alarm events, replay false-authorized-actuation mean, and replay false-authorized-actuation standard deviation.

Bootstrap confidence intervals use 1000 resamples of injected-fault trials, not individual samples. Paired Wilcoxon signed-rank tests compare AASVR-R against each baseline on per-trial balanced accuracy and replay false-authorized actuation, with Holm correction applied separately across the 13-baseline balanced-accuracy family and the 13-baseline replay-authorization family. Rank-biserial correlation and paired d_z are reported in the reproducibility tables, and a sign test is also reported for replay false-authorized-actuation differences because that distribution is zero-inflated and skewed. Sensor–fault grids use Benjamini–Hochberg correction.

7. Results

The results focus on two claims: replay detection is competitive with tuned baselines, and lower replay false-authorized actuation comes with a missed-authorization cost that must be reported in the same place.

7.1. Hydroponic deployment characterization

The deployment data are first used to characterize the real measurement background on which AASVR-R operates. The 121244-row feed covers a continuous 30-day lettuce growth period, with approximately 16-second median sampling. The key point for this paper is not that the dashboard collected data, but that the closed-loop system generated the kind of sensor-to-actuator stream in which a bad reading can trigger a physical correction.

Table 5: Hydroponic measurement streams used in the primary AASVR-R analysis. The table separates deployed sensing from quantitative use so that the water-level exclusion is explicit rather than hidden.

Variable	Module	Primary analysis	Control relevance
EC	Solution	Yes	Nutrient/water dosing evidence
pH	Solution	Yes	Acid/base dosing evidence
Humidity	Growth	Yes	Growth-environment monitoring
Air temperature	Growth	Yes	Air-cooling evidence
Water temperature	Solution	Yes	Chiller evidence
CO2	Growth	Yes	CO2 valve evidence
Water level	Solution	No	Warning channel; calibration unresolved

7.2. Hydroponic synthetic-fault benchmark

Table 6 summarizes the primary target-domain result on 270 held-out synthetic-fault trials. With the validation-selected Page/CUSUM residual gate, AASVR-R remains close to the base AASVR detector on balanced accuracy while lowering replay false-authorized actuation from 0.196 to 0.041 events per trial. The same table shows the cost: AASVR-R's missed-authorization rate is 0.970, compared with 0.874 for base AASVR. AASVR-R matches LOF within bootstrap uncertainty while preserving higher specificity and fewer alerts. One-Class SVM is statistically similar to AASVR-R on replay false-authorized actuation, but it obtains this by rejecting aggressively, with a false-positive rate of 0.667 and nearly 890 alert samples per trial. The result is therefore a replay tradeoff, not a closed-loop control-performance claim.

The conservative operating point is chosen for the hydroponic dosing setting, not because it is universally optimal. In the nutrient-solution loop, an erroneous pH or EC dose enters the recirculating root zone and can require dilution, counter-dosing, and another mixing interval before the state is trustworthy again [26, 19]. By contrast, a delayed correction is partly bounded by continuous monitoring and by the 10-minute pH/EC lockout already imposed after dosing. The selected AASVR-R setting therefore prioritizes avoiding untrusted dosing authorization over responsiveness, which is appropriate for slow NFT solution management but would need retuning for faster or safety-critical control loops.

The base-AASVR comparison is a three-way replay tradeoff. AASVR-R has a small but highly consistent balanced-accuracy decrement relative to base AASVR (ΔBA $= -0.001$, standard deviation about 0.0018), a higher

Table 6: Held-out hydroponic synthetic-fault benchmark. Baselines are validation-tuned under the corrected objective in Eq. (2) and evaluated on the same test split, so values may differ from earlier internal reports. Missed authorization is reported only for AASVR variants because it depends on their trusted-state unsafe-band trace; lower is better for replay false-authorized actuation and missed authorization.

Method	Recall	Specificity	Bal. acc.	Alerts	Replay false auth.	SD	Missed auth.
AASVR	0.821	0.870	**0.845**	79.4	0.196	0.738	0.874
AASVR-R	0.821	0.867	0.844	79.4	**0.041**	**0.290**	0.970
LOF	0.853	0.798	0.826	276.4	0.130	0.629	–
Kalman	0.706	0.763	0.735	320.6	0.404	1.061	–
Isolation Forest	0.581	0.826	0.703	236.0	0.630	1.277	–
Hampel	0.411	0.984	0.698	24.4	1.041	1.502	–
Moving median	0.411	0.984	0.698	24.4	0.993	1.496	–
Recursive PCA	0.317	0.988	0.652	17.9	1.319	1.653	–
One-Class SVM	**0.912**	0.333	0.623	889.7	0.070	0.403	–
GLR	0.129	**0.999**	0.564	3.3	1.330	1.638	–

Table 7: Paired held-out effects for AASVR-R against representative methods. Positive ΔBA means AASVR-R has higher per-trial balanced accuracy; positive Δreplay false authorization means the comparison method has more replay false-authorized actuations than AASVR-R. d_z is the paired standardized BA effect size; $p_{\mathrm{FA,sign}}$ is Holm-corrected for the sign-test replay-authorization family.

Baseline	ΔBA	d_z	p_{BA}	ΔReplay false auth.	95% CI	$p_{\mathrm{FA,sign}}$
Base AASVR	-0.001	-0.719	1.000	0.156	[0.081, 0.241]	< 0.001
LOF	0.018	0.113	< 0.001	0.089	[0.030, 0.152]	0.021
One-Class SVM	0.221	1.240	< 0.001	0.030	[-0.026, 0.081]	0.212
Kalman	0.110	0.552	< 0.001	0.363	[0.230, 0.496]	< 0.001
Hampel	0.147	0.646	< 0.001	1.000	[0.822, 1.185]	< 0.001
CUSUM	0.204	0.946	< 0.001	1.233	[1.044, 1.426]	< 0.001

missed-authorization rate, and a lower replay false-authorized-actuation count. The negative detection sign is therefore stated explicitly rather than treated as an advantage; the operational contrast is whether the replay false-authorized-actuation reduction is worth the additional authorization suppression. The LOF balanced-accuracy difference is statistically detectable but operationally small ($d_z = 0.113$). Its practical contrast is the control profile: AASVR-R has higher specificity, fewer alerts, and lower replay false-authorized actuation. For One-Class SVM, the replay-authorization contrast is not distinguishable; the separation is instead its aggressive rejection and low specificity.

Figure 3 expands this point across authorization operating settings. The grid varies trust threshold, authorization persistence, replay cooldown, and response-reliability risk thresholds while keeping the held-out fault windows fixed. The manuscript setting is intentionally conservative: it sits near the low replay-false-authorized-actuation region, but it does so by accepting a high missed-authorization rate. Less conservative Pareto settings are available in the sweep when the deployment objective is tighter real-time regulation rather than high-confidence interlock behavior. This figure is included to make the operating tradeoff explicit rather than optimizing only for one favorable error direction.

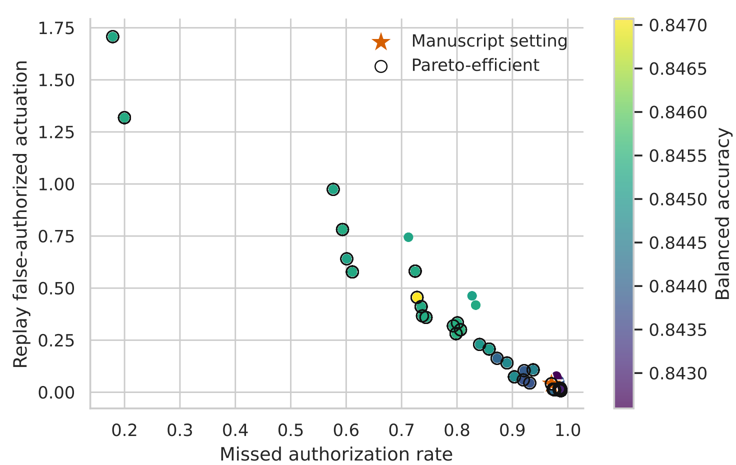

Figure 3: Replay authorization operating tradeoff across AASVR-R settings on the held-out hydroponic synthetic-fault split. Each point is one authorization setting; color shows balanced accuracy. The starred point is the manuscript setting.

The reproducibility artifact also writes a sample audit trail for a held-out pH replay. In that replay, a pH jump above 8.0 is logged as a trusted-deviation failure, the trusted pH remains near 6.09, the state moves through suspect and persistent-deviation states, and no actuation is authorized from the untrusted value.

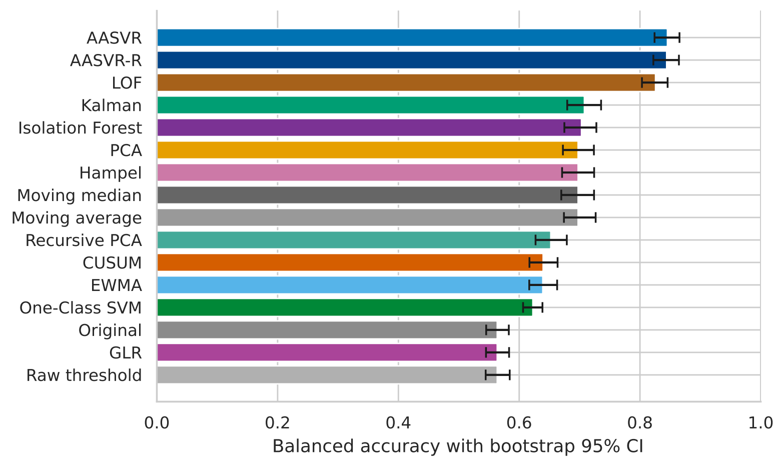

Figure 4: Bootstrap uncertainty for balanced accuracy. Intervals are trial-level resamples.

7.3. Sensor-wise and fault-type comparisons

Table 8 breaks down AASVR-R performance by hydroponic sensor. The CUSUM residual gate substantially improves the EC stream relative to the earlier single-window gate, so all primary sensors now remain above 0.83 balanced accuracy. CO2 remains the strongest stream. EC and pH are still the most operationally sensitive because dosing actions can change the nutrient solution, but the response-memory layer keeps their false-authorized actuation low in replay. These per-sensor and per-fault rows contain 54 and 30 held-out trials respectively, so they are descriptive breakdowns rather than as stable as the aggregate bootstrap comparison.

Table 9 shows performance by injected fault class. AASVR-R performs best on spike, saturation, dropout, step, bias, noise-burst, and multi-spike faults. The sequential CUSUM residual gate improves slow-shift and EC-bias behavior but does not solve frozen plausible values. Stuck-at remains at or below chance because an exactly repeated value can be indistinguishable from a genuinely steady process unless there is independent excitation, reference instrumentation, or a command-response episode.

Table 8: AASVR-R synthetic-fault performance by hydroponic sensor. Each sensor row contains 54 held-out trials; replay false authorization is reported as mean events per injected-fault trial.

Sensor	Recall	Specificity	Bal. acc.	Replay false auth.
CO2	**0.854**	**0.878**	**0.866**	0.056
Water temperature	0.821	0.870	0.846	0.037
Air temperature	0.823	0.860	0.841	0.093
pH	0.808	0.863	0.836	0.019
EC	0.801	0.863	0.832	**0.000**

Table 9: AASVR-R synthetic-fault performance by injected fault type. Each fault-type row contains 30 held-out trials. Stuck-at and drift are the most difficult conditions because frozen plausible values and slow monotonic movement can resemble real process behavior.

Fault type	Recall	Specificity	Bal. acc.	Replay false auth.
Spike	0.996	**0.875**	**0.935**	**0.000**
Saturation	**1.000**	0.862	0.931	**0.000**
Dropout	**1.000**	0.854	0.927	**0.000**
Step	0.976	**0.875**	0.925	**0.000**
Bias	0.983	0.866	0.925	**0.000**
Noise burst	0.936	0.873	0.904	0.033
Multi-spike	0.932	0.867	0.900	**0.000**
Drift	0.496	0.861	0.678	0.067
Stuck-at	0.074	0.869	0.472	0.267

Table 10: Diagnostic-excitation replay for plausible stuck or weak-response conditions. Values are averaged over pH, EC, humidity, air temperature, water temperature, and CO2. The replay uses bounded diagnostic pulses as implementation evidence only; measured actuator logs would be required to treat these rows as physical deployment evidence.

Diagnostic scenario	Correct decision	Fault detected	Alert	Delay (samples)	Repeated auth.
Healthy diagnostic pulse	**1.000**	–	0.000	–	**0.000**
Stuck during diagnostic	**1.000**	**1.000**	**1.000**	12.0	**0.000**
Weak diagnostic response	**1.000**	**1.000**	**1.000**	12.0	**0.000**

Table 10 shows the implemented mitigation pathway for the stuck-at gap. A periodic diagnostic excitation pulse can expose a frozen plausible value when the measured process fails to move in the expected direction. Such pulses should be enabled only under conservative conditions: the trusted state is in band, no lockout or manual override is active, the diagnostic runtime or dose is bounded, and the post-command response window is logged. Without such an episode, an independent reference sample, or a real actuator-response record, AASVR-R still cannot guarantee detection of a plausible frozen value.

7.4. Component contribution and ablation

Table 11 separates the contribution of the supervisory components on the same 270 held-out test trials as Table 6. Lockout-only control has no fault detection and still produces 1.356 replay false-authorized actuations per trial, so the low AASVR-R replay-authorization rate is not explained by the existing confirmation/cooldown rule alone. The table also shows the central tradeoff: reducing replay false-authorized actuation by adding response reliability increases the missed-authorization rate relative to base AASVR. The trust score is retained as an audit and authorization diagnostic, not as an independent detector contribution; in the selected configuration, hard gates and persistence dominate until the score is coupled to the full authorization logic.

Table 11: Component-contribution analysis on the full 270-trial held-out synthetic-fault test split. Values are mean events or scores per trial.

Variant	Recall	Specificity	Bal. acc.	Replay false auth.	Missed auth.
Lockout only	0.000	1.000	0.500	1.356	0.850
Raw threshold + supervisor	0.129	**0.999**	0.564	1.330	–
Robust gates + hold	0.737	0.887	0.812	0.363	0.873
Full AASVR	0.821	0.870	**0.845**	0.196	0.874
Full AASVR-R	0.821	0.867	0.844	**0.041**	0.970
AASVR without cooldown	0.821	0.870	**0.845**	0.411	0.728
AASVR without confirmation/cooldown	0.821	0.869	**0.845**	1.730	0.152

Internal ablations and sensitivity sweeps are retained in the reproducibility tables rather than expanded in the main text. They show that removing response reliability restores the base-AASVR replay false-authorized-actuation rate (0.196), removing cooldown raises replay false-authorized actuation to 0.411, and removing actuation confirmation plus cooldown raises it to 1.730. The corrected validation objective selects the same reported setting, $c_s = 6.0$, $q_{\min} = 0.70$, and transient limit 2, under balanced-accuracy-only, reported, high replay-authorization, and high missed-authorization profiles. Focused sweeps over (α_c, h_c) and response-memory decay show neighboring

CUSUM settings within 0.001 balanced accuracy of the manuscript setting and confirm that $\lambda = 0.50$ gives the lowest replay false-authorized-actuation rate among the tested response-memory settings.

7.5. Deployment-context checks

The full-feed deployment replay and agronomic records are retained as context rather than evidence of closed-loop superiority. The P1 hydroponic crop cycle produced the sensor-to-actuator decision stream; P2 and P3 did not, and treatment differences are confounded with enclosure, lighting, cooling, CO_2 regulation, nutrient delivery, and automation. On the full P1 replay, AASVR-R authorizes only 11 sensor-sample decisions, compared with 60723 for base AASVR and 192506 for raw thresholding, while the missed-authorization opportunity rate is 0.788. This is replay silence, not measured agronomic or dosing improvement. It means the selected configuration behaves as a high-confidence interlock for low-frequency dosing rather than as a replacement controller; a deployment that needs tighter regulation should start from one of the less conservative operating points in Fig. 3. Counterfactual response and diagnostic-excitation outputs are likewise retained as implementation checks unless independent actuator-command/state records are supplied. The repository now includes the reconstruction and scoring path for those records, so a controller export can add measured command-state agreement and post-command directional-response results without changing the synthetic-fault benchmark.

8. Discussion

8.1. Why actuation awareness matters

The component, ablation, and audit outputs show why actuation-aware validation should be evaluated with both error directions and decision transparency. After adding the validation-selected Page/CUSUM residual gate, AASVR-R matches the strongest non-AASVR detector within bootstrap uncertainty while keeping fewer alerts than LOF and lower replay false-authorized actuation than base AASVR. The detector gain comes from a standard sequential residual test rather than from a learned one-class boundary, so the novelty is not the CUSUM detector itself but its placement inside an auditable measurement-to-actuation supervisor. The missed-authorization rates show the cost of conservative authorization and prevent the replay false-authorized-actuation result from being read as a one-sided win. For hydroponic systems, these distinctions are practical because a logged anomaly, a held trusted state, a pump command, a diagnostic pulse, and a failed post-command response are different engineering decisions.

8.2. External transfer

The supplementary HAI and SKAB checks show that the same streaming interface can run on independent process datasets, but direct parameter transfer is weak. HAI attacks and SKAB process anomalies are not identical to hydroponic sensor faults, and the current method uses hydroponic physical ranges and rate assumptions. These results are therefore not used as evidence of cross-domain dominance; they identify the need for process-specific limits or model residuals before claiming industrial transfer.

8.3. Limitations

The hydroponic experiment does not provide independent reference instruments for every sensor at every time point. Synthetic fault injection is therefore necessary for controlled scoring, but injected faults cannot cover every real failure mode. The reported replay false-authorized-actuation values are replayed authorization events, not measured physical process disturbances, because the replay benchmark does not close the loop around real actuators. The historical deployment also did not retain timestamped actuator-command/state columns with relay state, command source, and lockout state. The response and diagnostic-excitation replays are therefore counterfactual implementation checks: replay-derived authorizations and safe diagnostic pulses stand in for actuator commands because measured command-response evidence is unavailable.

The remaining limitations are therefore identifiable and measurable. Stuck-at behavior and weak response remain the principal algorithmic weaknesses because they can resemble real steady nutrient-solution dynamics without reference instrumentation or measured actuator effects. AASVR-R reduces repeated authority after failed response and improves slow-shift detection through the CUSUM residual gate, and Table 10 shows how bounded diagnostic excitation can expose plausible frozen values through missing response evidence. It still does not make frozen plausible values fully detectable when no command-response, diagnostic episode, or independent reference sample occurs. Water level is excluded from primary analysis because the available raw values are not defensibly calibrated; the repository now includes a calibration schema and fitting script that should be satisfied before reintegration. Crop outcomes can be linked to replay-derived exposure metrics as full-cycle deployment context, but they remain non-causal because enclosure, lighting, cooling, nutrient delivery, carbon dioxide regulation, and automation are confounded.

9. Conclusion

We presented a closed hydroponic control system and proposed AASVR-R as an auditable sensor-validation and actuation-authorization interface for that setting. The method separates validation, state update, rectification, actuation authorization, and response verification so that untrusted measurements can be logged without automatically driving physical commands. On held-out hydroponic synthetic faults, the standard CUSUM residual gate makes AASVR-R competitive with tuned detectors, while the response-aware supervisor lowers replay false-authorized actuation at the cost of a higher missed-authorization rate. The remaining requirement is not another passive filter alone, but real actuator-state logs and reference measurements that make command-response evidence observable during future deployments.

Data Availability

All data and code used for this study are available in the GitHub repository disclosed in the submission metadata. Timestamped actuator-command/state evidence is not available for the historical deployment because those logs were not retained, so the replay false-authorized-actuation results should not be interpreted as measured physical disturbances.

Declaration of Competing Interest

The authors declare no competing interests for the current manuscript draft.

Declaration of Generative AI and AI-Assisted Technologies

AI-assisted tools were used to help organize code, draft text, and check consistency. The authors remain responsible for the scientific content, claims, results, and final manuscript.

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